

```

1  program matrix_multiplication
2  implicit none
3  integer :: r1, c1, r2, c2, i, j
4  real, allocatable :: A(:, :), B(:, :), C(:, :), C_verify(:, :)
5
6  ! Get matrix dimensions from the user
7  print *, "Enter number of rows and columns for Matrix A:"
8  read(*, *) r1, c1
9  print *, "Enter number of rows and columns for Matrix B:"
10 read(*, *) r2, c2
11
12 ! Check if matrix multiplication is possible
13 if (c1 /= r2) then
14 print *, "Matrix multiplication not possible. Number of columns in A must equal number of rows in B."
15 stop
16 endif
17
18 ! Allocate matrices based on input dimensions
19 allocate(A(r1, c1), B(r2, c2), C(r1, c2), C_verify(r1, c2))
20
21 ! Read matrices from file
22 open(unit=10, file="matrix_A1.txt")
23 do i = 1, r1
24 read(10, *) (A(i, j), j = 1, c1)
25 end do
26 close(10)
27
28 open(unit=20, file="matrix_B2.txt")
29 do i = 1, r2
30 read(20, *) (B(i, j), j = 1, c2)
31 end do
32 close(20)
33
34 ! Call subroutine to perform matrix multiplication
35 call matrix_multiply(A, r1, c1, B, r2, c2, C)
36
37 ! Verify the result using Fortran's matmul function
38 C_verify = matmul(A, B)
39
40 ! Print the resulting matrix C
41 print *, "Matrix C (result of A * B):"
42 do i = 1, r1
43 print *, (C(i, j), j = 1, c2)
44 end do
45
46 ! Print the verified matrix C_verify
47 print *, "Matrix C_verify (using matmul):"
48 do i = 1, r1
49 print *, (C_verify(i, j), j = 1, c2)
50 end do
51
52 ! Check if the manual multiplication matches matmul result
53 if (all(C == C_verify)) then
54 print *, "The result is verified and correct!"
55 else
56 print *, "There is a discrepancy between manual calculation and matmul."
57 endif
58
59 ! Deallocate arrays
60 deallocate(A, B, C, C_verify)
61
62 end program
63
64 ! Subroutine to perform matrix multiplication
65 subroutine matrix_multiply(A, r1, c1, B, r2, c2, C)
66 implicit none
67 integer :: r1, c1, r2, c2

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```
67  real :: A(r1, c1), B(r2, c2)
68  real :: C(r1, c2)
69  integer :: i, j, k
70
71  ! Initialize result matrix C to zero
72  C = 0.0
73
74  ! Perform matrix multiplication
75  do i = 1, r1
76  do j = 1, c2
77      do k = 1, c1
78          C(i, j) = C(i, j) + A(i, k) * B(k, j)
79      end do
80  end do
81  end do
82  end subroutine matrix_multiply
83
```